

```
"""
```

```
STEP 0: Install required libraries for RAG pipeline
```

```
This step ensures all required packages are available.  
Even if already installed in Colab, we include this for completeness.
```

```
"""
```

```
# Install LangChain (for building RAG pipeline)  
!pip install langchain
```

```
# Install ChromaDB (vector database)  
!pip install chromadb
```

```
# Install sentence-transformers (for embeddings)  
!pip install sentence-transformers
```

```
# Install OpenAI (for LLM)  
!pip install openai
```

```
Requirement already satisfied: langchain in /usr/local/lib/python3.12/dist-packages (1.2.14)  
Requirement already satisfied: langchain-core<2.0.0,>=1.2.10 in /usr/local/lib/python3.12/dist-packages (from langchain) (1.2.10)  
Requirement already satisfied: langgraph<1.2.0,>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from langchain) (1.1.4)  
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain) (2.12.3)  
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.10) (1.33.0)  
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.10) (0.3.45)  
Requirement already satisfied: packaging>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.10) (23.2.0)  
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.10) (6.0.1)  
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.10) (8.1.0)  
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.10) (4.7.0)  
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.10) (0.12.0)  
Requirement already satisfied: langgraph-checkpoint<5.0.0,>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from langgraph<1.2.0,>=1.1.1) (2.1.0)  
Requirement already satisfied: langgraph-prebuilt<1.1.0,>=1.0.8 in /usr/local/lib/python3.12/dist-packages (from langgraph<1.2.0,>=1.1.1) (1.0.8)  
Requirement already satisfied: langgraph-sdk<0.4.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from langgraph<1.2.0,>=1.1.1) (0.3.0)  
Requirement already satisfied: xxhash>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from langgraph<1.2.0,>=1.1.1->langgraph-prebuilt<1.1.0,>=1.0.8) (3.5.0)  
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4) (0.6.0)  
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4) (2.41.4)  
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4) (0.4.2)  
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0) (1.9)  
Requirement already satisfied: ormsgpack>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from langgraph-checkpoint<5.0.0,>=2.1.0) (1.12.0)  
Requirement already satisfied: httpx>=0.25.2 in /usr/local/lib/python3.12/dist-packages (from langgraph-sdk<0.4.0,>=0.3.0) (0.25.2)  
Requirement already satisfied: orjson>=3.11.5 in /usr/local/lib/python3.12/dist-packages (from langgraph-sdk<0.4.0,>=0.3.0) (3.11.5)  
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45) (1.0.0)  
Requirement already satisfied: requests>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45) (2.31.0)  
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45) (0.23.0)  
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0) (3.7.1)  
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0) (2024.7.4)  
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0) (1.0.5)  
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0) (3.10)  
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0) (0.14.0)  
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.0->langgraph-checkpoint<5.0.0,>=2.1.0) (3.4.0)  
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.0->langsmith<1.0.0,>=0.3.45) (2.2.3)  
Collecting chromadb  
  Downloading chromadb-1.5.7-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (5.0 kB)  
Collecting build>=1.0.3 (from chromadb)  
  Downloading build-1.4.2-py3-none-any.whl.metadata (5.8 kB)  
Requirement already satisfied: pydantic>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.12.3)  
Requirement already satisfied: pydantic-settings>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.13.1)  
Collecting pybase64>=1.4.1 (from chromadb)  
  Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl (10.5 kB)  
Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.30.3)  
Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.0.2)  
Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.15.0)  
Collecting onnxruntime>=1.14.1 (from chromadb)  
  Downloading onnxruntime-1.24.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.2 kB)  
Requirement already satisfied: opentelemetry-api>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.38.0)  
Collecting opentelemetry-exporter-otlp-proto-grpc>=1.2.0 (from chromadb)  
  Downloading opentelemetry_exporter_otlp_proto_grpc-1.40.0-py3-none-any.whl.metadata (2.6 kB)  
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.38.0)  
Requirement already satisfied: tokenizers>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.22.2)  
Collecting pypika>=0.48.9 (from chromadb)  
  Downloading pypika-0.51.1-py2.py3-none-any.whl.metadata (51 kB)  
Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.67.3)  
Requirement already satisfied: overrides>=7.3.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (7.7.0)  
Requirement already satisfied: importlib-resources in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.5.2)  
Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.80.0)
```

```
"""
```

```
Install updated LangChain ecosystem packages
"""
```

```
!pip install langchain langchain-community langchain-openai chromadb sentence-transformers langchain-huggingface -q
```

```
===== 2.5/2.5 MB 32.2 MB/s eta 0:00:00
===== 88.5/88.5 kB 4.8 MB/s eta 0:00:00
===== 1.0/1.0 MB 23.0 MB/s eta 0:00:00
===== 64.9/64.9 kB 4.1 MB/s eta 0:00:00
===== 51.0/51.0 kB 1.7 MB/s eta 0:00:00
```

```
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.33.1 which is incompatible.
google-adk 1.28.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you have opentelemetry-api 1.40.0 which is incompatible.
google-adk 1.28.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have opentelemetry-sdk 1.40.0 which is incompatible.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-common==1.38.0, but you have openteleme
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you have opentelemetry-proto 1.40.0 whic
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you have opentelemetry-sdk 1.40.0 which is
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0,>=1.35.0, but you have opentelemetry-sdk 1.40.0 wh
```

```
from langchain_huggingface.embeddings import HuggingFaceEmbeddings
```

```
"""
```

```
STEP 2: Load embedding model
```

```
Embeddings convert text into numerical vectors.
```

```
We use a pre-trained model: all-MiniLM-L6-v2
```

```
"""
```

```
# Initialize embedding model
```

```
embedding_model = HuggingFaceEmbeddings(
    model_name="all-MiniLM-L6-v2"
)
```

```
print("Embedding model loaded successfully")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
```

```
The secret `HF_TOKEN` does not exist in your Colab secrets.
```

```
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
```

```
You will be able to reuse this secret in all of your notebooks.
```

```
Please note that authentication is recommended but still optional to access public models or datasets.
```

```
warnings.warn(
```

```
modules.json: 100% 349/349 [00:00<00:00, 35.5kB/s]
```

```
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 10.3kB/s]
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
```

```
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
```

```
README.md: 10.5k/? [00:00<00:00, 674kB/s]
```

```
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.29kB/s]
```

```
config.json: 100% 612/612 [00:00<00:00, 48.1kB/s]
```

```
model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 454MB/s]
```

```
Loading weights: 100% 103/103 [00:00<00:00, 438.68it/s, Materializing param=pooler.dense.weight]
```

```
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
```

```
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |
```

```
Notes:
```

```
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
tokenizer_config.json: 100% 350/350 [00:00<00:00, 27.4kB/s]
```

```
vocab.txt: 232k/? [00:00<00:00, 10.3MB/s]
```

```
tokenizer.json: 466k/? [00:00<00:00, 19.4MB/s]
```

```
special_tokens_map.json: 100% 112/112 [00:00<00:00, 5.25kB/s]
```

```
config.json: 100% 190/190 [00:00<00:00, 18.2kB/s]
```

```
Embedding model loaded successfully
```

```
"""
```

```
Remove old/broken LangChain installations
```

```
"""
```

```
!pip uninstall -y langchain langchain-community langchain-openai
```

```
Found existing installation: langchain 1.2.14
Uninstalling langchain-1.2.14:
  Successfully uninstalled langchain-1.2.14
Found existing installation: langchain-community 0.4.1
Uninstalling langchain-community-0.4.1:
  Successfully uninstalled langchain-community-0.4.1
Found existing installation: langchain-openai 1.1.12
Uninstalling langchain-openai-1.1.12:
  Successfully uninstalled langchain-openai-1.1.12
```

```
"""
```

```
STEP 1: Completely remove conflicting packages
```

```
"""
```

```
!pip uninstall -y langchain langchain-core langchain-community langchain-openai langchain-text-splitters
```

```
WARNING: Skipping langchain as it is not installed.
Found existing installation: langchain-core 1.2.23
Uninstalling langchain-core-1.2.23:
  Successfully uninstalled langchain-core-1.2.23
WARNING: Skipping langchain-community as it is not installed.
WARNING: Skipping langchain-openai as it is not installed.
Found existing installation: langchain-text-splitters 1.1.1
Uninstalling langchain-text-splitters-1.1.1:
  Successfully uninstalled langchain-text-splitters-1.1.1
```

```
"""
```

```
Install required packages (compatible with current environment)
```

```
"""
```

```
!pip install langchain chromadb sentence-transformers -q
```

```
===== 112.7/112.7 kB 3.2 MB/s eta 0:00:00
===== 508.7/508.7 kB 12.1 MB/s eta 0:00:00
===== 169.8/169.8 kB 5.8 MB/s eta 0:00:00
```

```
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the
langchain-classic 1.0.3 requires langchain-text-splitters<2.0.0,>=1.1.1, which is not installed.
```

```
"""
```

```
Install missing dependency
```

```
"""
```

```
!pip install langchain-text-splitters -q
```

```
!pip install chromadb sentence-transformers -q
```

```
"""
```

```
Import required libraries
```

```
"""
```

```
from sentence_transformers import SentenceTransformer
import chromadb
```

```
"""
```

```
Load embedding model
```

```
"""
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
Loading weights: 100% 103/103 [00:00<00:00, 381.29it/s, Materializing param=pooler.dense.weight]
```

```
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
```

```
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |
```

```
Notes:
```

```
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
"""
```

```
Sample dataset
```

```
"""
```

```
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
```

```
"Deep Learning uses neural networks.",
"Natural Language Processing deals with text data.",
"RAG combines retrieval and generation."
]
```

```
"""
Create ChromaDB client and collection
"""

client = chromadb.Client()

collection = client.create_collection(name="rag_collection")
```

```
"""
Convert documents into embeddings and store
"""

for i, doc in enumerate(documents):
    embedding = model.encode(doc).tolist()

    collection.add(
        ids=[str(i)],
        documents=[doc],
        embeddings=[embedding]
    )
```

```
"""
Ask a query
"""

query = "What is Artificial Intelligence?"

query_embedding = model.encode(query).tolist()
```

```
"""
Retrieve most relevant documents
"""

results = collection.query(
    query_embeddings=[query_embedding],
    n_results=2
)

print("Retrieved Documents:")
for doc in results['documents'][0]:
    print("-", doc)

Retrieved Documents:
- Artificial Intelligence is the simulation of human intelligence.
- Machine Learning is a subset of AI.
```

```
"""
Generate answer (basic RAG output)
"""

context = " ".join(results['documents'][0])

answer = f"Based on retrieved data: {context}"

print("\nFinal Answer:")
print(answer)

Final Answer:
Based on retrieved data: Artificial Intelligence is the simulation of human intelligence. Machine Learning is a subset of AI.
```

